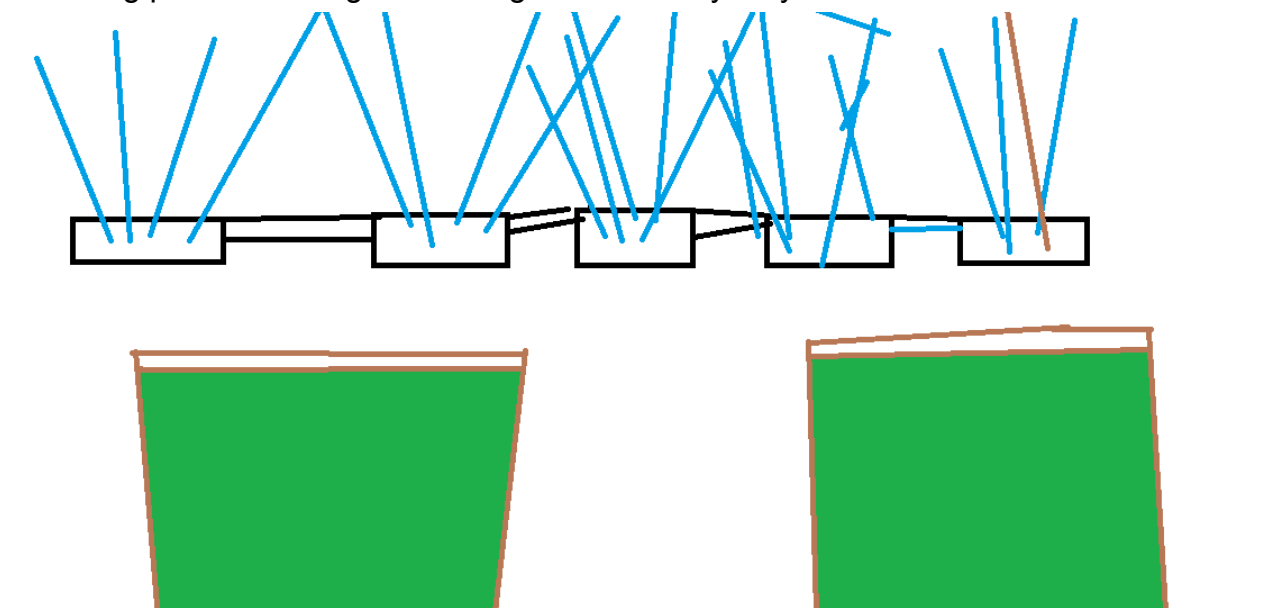


One prior solution was this multi sprinkler idea where we'd have 5 sprinklers (shown below in a line) with hoses attached to all of them so the water could easily run through all of them and set them all off at once. One reason this didn't work is because if it rained earlier in the day or the day before, the timer we put the sprinklers on would go off and drown the plants. And it would keep doing that until our stakeholder arrived back home after two weeks of being away, at that point the plants would be in pools and the damage would be irreversible. Another reason this wouldn't work is because we didn't have enough time to figure out how water would run through the mini hoses connecting the sprinklers and set them all off. This was a major issue and there were too many details and constraints we didn't take into account. On top of this, sprinklers are already used to fix this situation, we wouldn't actually be engineering anything "new," we'd just be taking pre made things and using them the way they're meant to be used in.



Another solution was a multi hose approach. Multiple hoses would line the inside garden beds and water all of them. A problem we ran into was the amount of hoses that would be used. We only had \$250 for our budget and hoses can get quite pricey. Along with that, the red parts seen in the picture below don't exist anywhere and we'd have to 3D print them, or make them out of PVC. That's a lot of microplastics and the durability of a 3D printed object wouldn't do too well on the outside for 2 weeks. Another thing is that if it rained our mechanism would still go off even if it downpoured the night before. Therefore drowning the plants and killing them. This idea also involved human interaction and if it broke it wouldn't be easily fixable. Our stakeholder wouldn't be home for two weeks and wouldn't be able to fix it if it broke. Another issue is that the mechanism could water the plants during the midday heat. This is very bad for the plants because it could literally burn them. We could've put a heat/sun sensor, but our build idea wasn't easily customizable.

